Death of a chimera: breath-synchronized collapse of a post-homoclinic transient in a two-population oscillator system

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Abstract

We report a long-duration empirical study of chimera collapse in the two-population Sakaguchi–Kuramoto system with strictly identical natural frequencies, on the oscillator engine of a deployed additive synthesizer at small population sizes ($N = 4$–64 per population). Its operational collapse detector — and the first-passage criterion standard for chimera lifetimes — measure the wrong observable: up to 98% of sustained near-synchrony crossings recover as the chimera re-forms, and 69–97% of detector firings are self-healing *grazes*. Under an absorption-grade label, true lifetimes at the operating corner ($A = 0.5$, $\beta = 0.05$) double, become N-independent ($\tau_{\mathrm{abs}} \approx 139\,\mathrm{s}$ over a $16\times$ range of N), and age (Weibull shape 2.3–3.0). Deaths are breath-synchronized (pooled Rayleigh $p = 2.2 \times 10^{-6}$) and ratchet: per-cycle envelope maxima grow monotonically to capture. The reduced two-population model of Abrams *et al.*, validated against its published bifurcations with no fitted parameters, explains the inversion: the operating corner lies beyond the homoclinic ($A_{hc}(0.05) = 0.4096 < 0.5$), so the deployed "chimera" is a transient spiraling out along a destroyed limit cycle's ghost, while the nominally unreliable corner ($A = 0.2$) holds the only true attractor ($r^* = 0.678$; 51–78% of seeds never absorb). It matches the breath period (25.0 vs. 21–25 s) and spiral rates and predicts individual lifetimes beyond any static feature, yet captures $3.2\times$ faster than the finite-N system dies. Finite-size fluctuations *prolong* the transient — an inversion of the noise-kills-metastability intuition; a direct test rules out additive collective noise, leaving the mechanism open.

Chimera states, in which a population of identical oscillators spontaneously splits into one synchronized group and one disordered group, are usually studied as a question in pure mathematics. Here the same phenomenon appears inside a working music synthesizer, where two coupled oscillator banks shape the sound and the software periodically judges the pattern "dead" and restarts it. Asking why it dies so often reveals that the restart trigger was watching the wrong quantity: most apparent deaths are momentary lapses that heal on their own. Once the genuine deaths are isolated, they prove regular, rhythmic, and no more frequent in large populations than small ones — and a simple reduced model explains where and why they occur, while underpredicting how long the pattern survives, a discrepancy the paper leaves as an open problem.

1 Introduction

Chimera states — the coexistence of synchronized and desynchronized populations in systems of identical coupled oscillators — have been a central object of nonlinear dynamics since their discovery by Kuramoto and Battogtokh [1] and naming by Abrams and Strogatz [2]. The two-population model of Abrams, Mirollo, Strogatz, and Wiley [3] made the phenomenon solvable: two identical populations with stronger intra- than inter-population coupling support stable chimeras, breathing chimeras born in a Hopf bifurcation, and a homoclinic boundary beyond which no chimera attractor survives. At finite size the picture is enriched by transience: on nonlocally coupled rings, chimeras are chaotic transients whose lifetimes grow with oscillator count [4], and the finite-N fate of chimeras has remained a recurring theme of the literature since [5, 6], including for two small populations of the kind studied here [7].

This paper studies chimera collapse in an unusual arena: the oscillator engine of a deployed real-time additive synthesizer, in which a two-population Sakaguchi–Kuramoto bank with *strictly identical* natural frequencies drives the partial amplitudes of a musical voice, and a supervisor process re-seeds the chimera whenever an engineering detector declares it dead. Coupled-oscillator networks have an established history as sound-synthesis drivers [8, 9]; here the synthesis application matters to the dynamics only in that it fixes the regime of interest — small populations ($N = 4$–64 per side), long unattended runtimes, and a specific operating corner chosen, as it turns out, for the wrong reason. The deployment prompts a question the theory literature rarely needs to ask operationally: *what exactly is the death of a chimera, and what does a detector of it actually detect?*

We answer with a chain of measurements, each gating the next, on a bit-reproducible integrator (2,100 baseline runs plus targeted re-campaigns; every figure regenerable from committed configurations). The contributions are:

1. **Graze versus absorption.** The operational collapse criterion — sustained near-synchrony of the incoherent population — overwhelmingly fires on events the chimera survives: up to 98% of criterion-satisfying crossings recover, and 69–97% of the deployed detector's firings are such grazes. We introduce a two-timescale, absorption-grade labeling and show first-passage "lifetimes" are roughly half the true absorption times (Sec. 3).

2. **An inverted stability picture.** Under absorption labeling, lifetimes at the operating corner are N-independent ($\approx 139\,\mathrm{s}$, $N = 4$–64, every seed absorbing), while at the nominally unreliable high-morph corner the majority of seeds *never* absorb: the operating corner has no chimera attractor at all, and the discarded corner holds the only persistent chimera in the system (Sec. 4).

3. **The death mechanism.** Collapse is a breath-synchronized ratchet: collapse attempts and true deaths lock to a phase of the breathing cycle, successive per-cycle envelope maxima grow monotonically (spiraling out exponentially toward the synchronization boundary), and hazard rises with cycle index. Transverse escape from the Poisson submanifold is ruled out by a validated observable and a null interventional test; lifetime is instead organized by the collective initial condition, dominantly the inter-population phase offset (Sec. 5).

4. **Closure by the reduced model — and a clean open problem.** The reduced equations of Ref. [3], validated against that paper's own published results and applied with a literal 1:1 timescale and zero fitted parameters, place the operating corner beyond the homoclinic ($A_{hc}(\beta{=}0.05) = 0.4096$), reproduce the breath period and per-cycle spiral rates, identify the never-absorbers' level with the stable chimera's r^*, and predict individual run lifetimes beyond any static feature of the seed. They also fail in one instructive way: mean-field capture is $3.2\times$ faster than finite-N death, and the flat $\tau_{\mathrm{abs}}(N)$ is absent from the N-free collective flow — finite-size fluctuations *extend* the transient (Sec. 6, Sec. 7).

2 System

The voice integrates two populations $\sigma \in \{1, 2\}$ of N phase oscillators each,

$$\dot{\theta}_i^{\sigma} \;=\; \omega \;+\; \sum_{\sigma'} \frac{K_{\sigma\sigma'}}{N} \sum_{j=1}^{N} \sin\!\big(\theta_j^{\sigma'} - \theta_i^{\sigma} - \alpha\big), \tag{1}$$

with intra-population coupling $K_{\sigma\sigma} = \mu = (1 + A)/2$, inter-population coupling $K_{\sigma\sigma'} = \nu = (1 - A)/2$ ($\mu + \nu = 1$, $A = \mu - \nu$), and phase lag $\beta = \pi/2 - \alpha$ [10, 11], matching the conventions of Ref. [3]. A code audit confirms the natural frequencies are identical *exactly* ($\delta\omega = 0$: no per-oscillator frequency term, detuning explicitly zeroed on this path), so the dynamics is deterministic given the initial condition and collapse cannot be attributed to frequency disorder; we work in the $\omega = 0$ rotating frame. Integration is fourth-order Runge–Kutta with $dt = 0.05$ at a documented one-model-unit-per-second clock, so all times below are seconds with no conversion; halving and quartering dt leaves lifetime distributions statistically indistinguishable (two-sample KS $p = 1.0$). All runs are bit-reproducible from logged (seed, parameters, dt); every re-analysis below begins by reproducing the upstream labels exactly.

In the application, the incoherent population's per-oscillator coherences modulate the gains of an additive partial bank, $g_i' = g_i \left[\, 1 + d\, a\, (c_i - \tfrac{1}{2}) \,\right]$ with $c_i = \tfrac{1}{2}(1 + \cos(\theta_i - \psi))$, turning the chimera's collective dynamics into a slowly self-morphing spectrum; the synthesis layer plays no further role in the dynamics studied here. Runs are initialized with the canonical chimera seed (one population tightly clustered, the other spread), since spontaneous chimera formation from random phases is negligible in this regime. The deployed supervisor declares the state alive while $\max_\sigma R_\sigma > 0.9$ and $\min_\sigma R_\sigma < 0.85$, fires after 2 s of sustained violation, and re-seeds the incoherent population. The baseline campaign comprises 200 seeds per population size at the operating corner ($A = 0.5$) and 100 at the high-morph corner ($A = 0.2$), $N \in \{4, 8, 16, 24, 32, 48, 64\}$, $\beta = 0.05$ throughout, integrated unsupervised to $t_{\max} = 2000$ s.

3 Graze versus absorption

What counts as the death of a chimera? In a deployed system the answer is operational: the supervisor's aliveness condition fails continuously for 2 s, and our initial lifetime campaign used the closely related first-passage criterion that $\min(R_1, R_2)$ exceed a threshold $\theta = 0.85$ sustained for $W = 5$ s. Both definitions encode the same assumption: that a sustained excursion of the incoherent population to near-synchrony *is* the death of the chimera. This section shows that assumption is wrong, and replaces it.

3.1 Threshold crossings overwhelmingly recover

Extending the recorded trajectories past their first criterion-satisfying crossing, rather than terminating there, reveals that the crossing is usually not absorbing. At the operating corner and across $N = 8$–64, up to 98% of first crossings — excursions exceeding θ for longer than the full 5 s sustain window — are followed by recovery: the incoherent population desynchronizes again, with post-crossing minima of $\min(R_1, R_2)$ clustering at 0.2–0.45, a fully re-formed chimera. About two-thirds of these recovered trajectories are later absorbed for good; the remainder churn through further graze–recover cycles. The first sustained crossing is therefore a *graze* of the synchronization boundary, not a death, and first-passage lifetimes computed from it measure time-to-first-long-graze.

Figure 1 shows a representative trajectory ($N = 16$, $A = 0.5$; seed pinned in the released configuration). The run grazes twice — each time exceeding θ for more than 5 s before the chimera re-forms — and is absorbed on its third approach at $t_{\text{abs}} = 221.9$ s, having first "collapsed" by the engineering criterion at $t_{\text{graze}} = 83.2$ s. The shipped supervisor's detector fires *six times* on this trajectory before the single event that is irreversible: six firings, two reforms, one death.

3.2 An absorption-grade label

We therefore assign every run two timescales from a single trace. The first, t_{graze}, is the unmodified first-passage label, retained for comparability. The second, t_{abs}, is the time of the first crossing *not* followed by

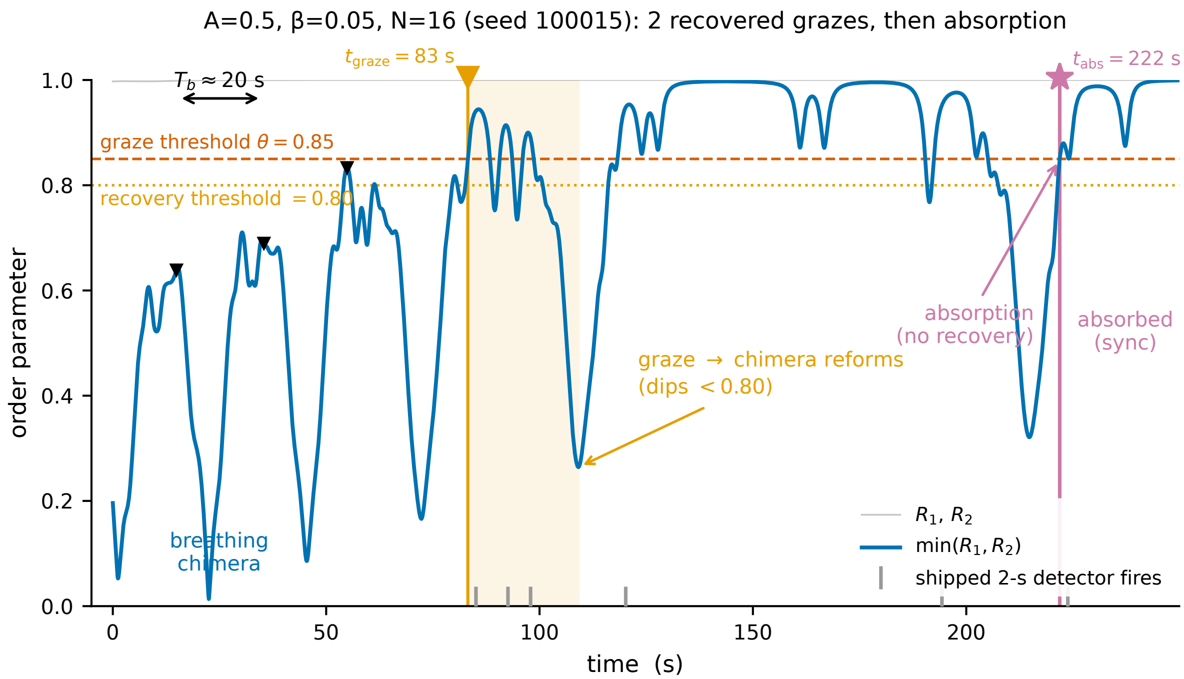

Figure 1: A graze is not a death. $\min(R_1, R_2)$ for a single run ($N = 16$, $A = 0.5$, $\beta = 0.05$; breath period $T_b = 20.5$ s). The trajectory crosses $\theta = 0.85$ (upper line) in sustained excursions twice, recovering below the 0.80 recovery threshold (lower line) each time as the chimera re-forms, before the absorbing crossing at $t_{\mathrm{abs}} = 221.9$ s. The engineering first-passage label $t_{\mathrm{graze}} = 83.2$ s and the six firings of the deployed 2-s detector are marked.

recovery — $\min(R_1, R_2) < 0.80$ sustained at least 5 s — within a verification horizon $T_v = 120$ s; runs still alternating between grazes and recoveries at the integration horizon are right-censored.

The absorption label is robust to its own knobs where it matters. Sweeping $T_v \in \{60, 120, 240\}$ s against recovery thresholds $\{0.75, 0.80\}$ moves $\hat{\tau}_{\mathrm{abs}}$ by 22–27% at $A = 0.5$, monotonically in T_v and negligibly in the recovery threshold — comparable to or better than the $\sim$52% (θ, W)-sensitivity of the graze criterion itself. At $A = 0.2$ the apparent sensitivity is dominated by censoring rather than the labeling parameters, for a reason Sec. 4 shows is dynamical: most $A = 0.2$ trajectories never absorb at all.

The quantitative consequence at the operating corner is a factor of two: $\hat{\tau}_{\mathrm{graze}} \approx 63$–$67$ s against $\hat{\tau}_{\mathrm{abs}} \approx 139$ s (Fig. 2). First-passage chimera lifetimes computed with engineering-style criteria systematically measure the wrong, and roughly halved, quantity.

3.3 The deployed detector is a graze detector

Replaying the shipped detector over the same unsupervised trajectories and classifying each firing by whether the underlying event self-recovers gives Table 1: 69–79% of firings at $A = 0.5$, and 95–97% at $A = 0.2$, occur on grazes the chimera survives on its own. The deployed system has been re-seeding a state that was mostly not dead — harmless in practice, since re-seeding lands near where the trajectory already was, but diagnostic of the conceptual gap this section closes: engineering collapse detection and dynamical absorption are different observables. All statistics below are computed on t_{abs} unless noted.

Table 1: Supervisor over-trigger: fraction of the deployed 2-s detector's firings, replayed over unsupervised traces, occurring on self-recovering grazes.

A	N	firings/run	over-trigger	median wasted (s)
0.5	8	7.2	79%	4.4
0.5	16	5.2	76%	2.4
0.5	32	4.1	73%	3.9
0.5	64	3.3	69%	5.3
0.2	8	12.3	95%	2.0
0.2	16	9.8	97%	1.8
0.2	32	7.9	96%	1.7
0.2	64	4.7	95%	1.8

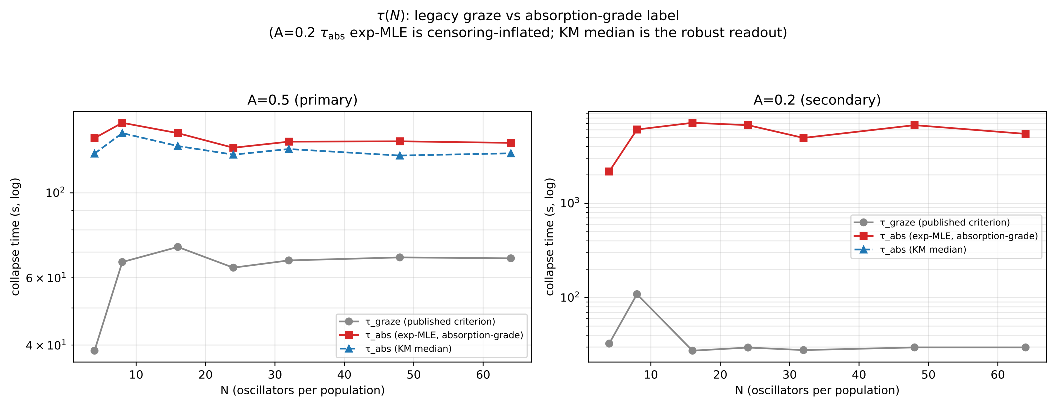

Figure 2: Graze versus absorption lifetimes. Exponential-MLE $\hat{\tau}_{\text{graze}}$ and $\hat{\tau}_{\text{abs}}$ versus N at both coupling disparities. At $A = 0.5$ absorption lifetimes are $\approx 2.2\times$ the first-passage values and flat in N; at $A = 0.2$ no absorption lifetime is defined for the majority of seeds (Sec. 4).

4 Absorption statistics at finite N

4.1 A flat plateau at the operating corner

At $A = 0.5$, every one of the 1,400 seeds absorbs within the horizon (0% censored at every N), and the absorption lifetime is N-independent: $\hat{\tau}_{\text{abs}} = 139, 152, 143, 131, 136, 136, 135$ s for $N = 4$–64, a ratio $\tau(64)/\tau(4) = 0.97$ over a sixteen-fold range of population size, with fitted exponential rate in N indistinguishable from zero. The weak rise from $N = 4$ to 16 visible in the graze times ($39 \to 72$ s) does not survive absorption labeling — it was a graze artifact. The published-style picture of a lifetime that grows with oscillator count, familiar from ring-topology chimeras [4], is absent here in both label and shape: adding partials does not buy persistence at this corner, at any tested N.

4.2 The inversion: the high-morph corner never lets go

The corner the application had classified as unreliable behaves in the opposite way. At $A = 0.2$, between 51% and 78% of seeds (rising with N) *never absorb*: extending the horizon eightfold to $t_{\max} = 16{,}000$ s leaves the censored fraction essentially unchanged, so the censoring is irreducible — a dynamical finding, not a horizon artifact. The minority that does absorb does so promptly and increasingly on the first approach:

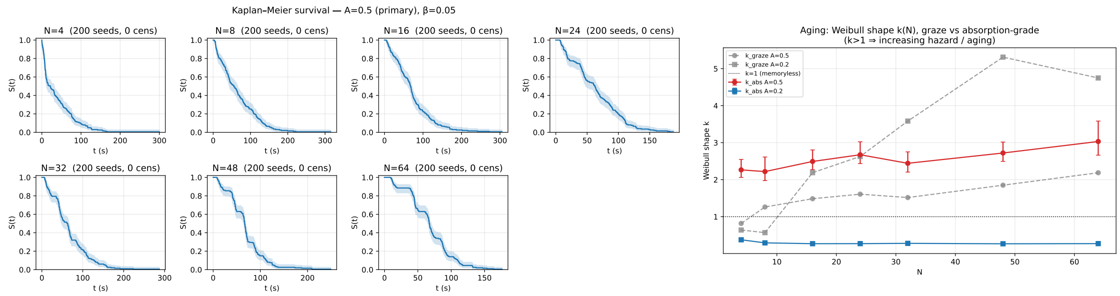

Figure 3: Absorption survival and aging. Kaplan–Meier survival curves by N (left, $A = 0.5$) develop a shoulder with increasing N; censored-Weibull shape $k_{\text{abs}}(N)$ (right) rises from 2.3 to 3.0 with all confidence intervals above the memoryless value $k = 1$.

the zero-graze fraction among absorbers rises from 0.08 at $N = 4$ to 0.96 at $N = 64$. Capture at $A = 0.2$ is bimodal — absorb almost immediately, or settle into a persistent, intermittently grazing chimera [12] with no defined absorption time. The stability folklore of the deployment is therefore inverted: the "reliable" corner always dies, and the "risky" corner contains the only immortal chimeras in the system. Section 6 shows both facts are exactly where the bifurcation diagram says they should be.

4.3 Aging: hazard rises with every breath

Absorption at $A = 0.5$ is not memoryless. Censored-Weibull fits give shape parameters k_{abs} of 2.27, 2.22, 2.49, 2.67, 2.44, 2.72, and 3.03 for the seven sizes $N = 4$–64, all confidence intervals well above 1 (Fig. 3) — stronger aging than in the graze times ($k_{\text{graze}} = 0.82$–2.19), so the survival shoulder is not a labeling artifact. The natural per-breath Bernoulli model — a constant absorption probability per high-R pass — is rejected at every $A = 0.5$ point (χ^2 $p < 0.001$ against geometric cycles-to-absorption; Weibull-in-cycles shape $k_{\text{cyc}} \approx 2.1$–2.9): the per-pass hazard *rises* with cycle index. The average per-pass absorption probability $\hat{p}$ nonetheless tracks the plateau, rising from 0.28 at $N = 4$ to 0.44 by $N = 24$ and flattening (Fig. 4), while the mean number of recovered grazes before death falls from 2.5 to 1.25. At $A = 0.2$ the strong apparent aging in the graze statistics (k_{graze} up to 5.3) is retired: on the few true absorptions $k_{\text{abs}} \approx 0.3 < 1$, a fast, front-loaded minority rather than aging. The never-absorbers hold a mean incoherent-population level $\overline{R}_{\text{incoh}} \approx 0.66$–0.68, sitting at the stable fixed point $r^* = 0.678$ identified in Sec. 6.

5 Mechanism: the breath-synchronized ratchet

5.1 Deaths lock to the breathing phase

The chimera at $A = 0.5$ breathes: $\min(R_1, R_2)$ oscillates with period $T_b \approx 21$–25 s, its envelope reaching ≈ 0.83 against the $\theta = 0.85$ boundary — the cycle passes within a hair of near-synchrony once per breath. Collapse *attempts* (long grazes) lock to this cycle: their phases relative to the preceding envelope peak cluster at 0.13–0.20 of the cycle, with resultant length tightening from 0.43 to 0.69 as N grows and pooled Rayleigh $p \approx 0$. True absorptions inherit the lock: pooled over $A = 0.5$, $\bar{R} = 0.21$ with Rayleigh $p = 2.2 \times 10^{-6}$ (three of six individually testable points significant; clustering weakens at large N, where capture completes within 2–3 cycles and leaves few phases to test). Mean absorption takes roughly six breaths ($\tau_{\text{abs}}/T_b \approx 6$). Death, when it comes, comes at a particular moment of the breath (Fig. 5).

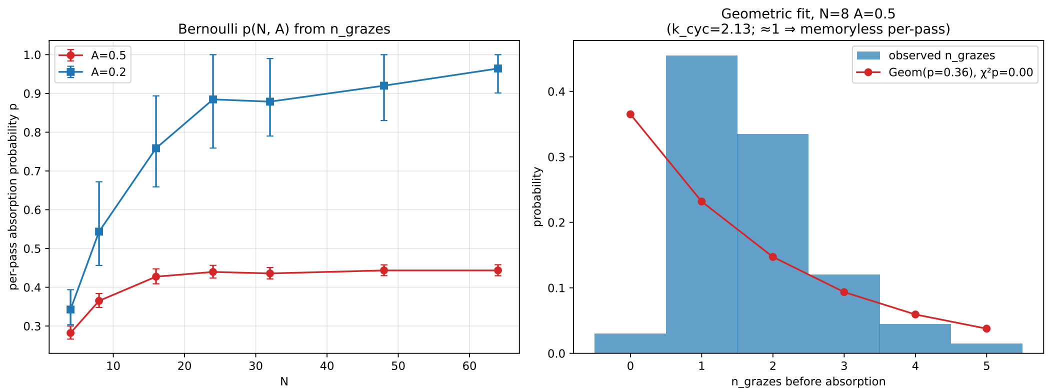

Figure 4: Per-pass absorption statistics at $A = 0.5$. Mean per-pass absorption probability $\hat{p}(N)$ rises then plateaus, mirroring $\tau_{\text{abs}}(N)$; the geometric (constant-p) model is rejected at every point — hazard rises with cycle index ($k_{\text{cyc}} > 1$).

5.2 The envelope ratchets monotonically to capture

Within individual runs, successive breathing maxima M_k of $\min(R_1, R_2)$ do not fluctuate about a level — they climb. The mean per-cycle increment $\langle \Delta M \rangle$ is $+0.031, +0.052, +0.078, +0.091$ for $N = 8, 16, 32, 64$, with cluster-bootstrap confidence intervals strictly positive at every point; the fraction of runs with monotone (allowing one inversion) maxima reaches 1.00 by $N = 32$. Where enough sub-threshold approach peaks exist to fit, $\log(\theta - M_k)$ falls linearly in cycle index with median slopes -0.55 to -0.77 and a negative slope in 100% of fittable runs: the approach to the synchronization boundary is an exponential spiral-out (Fig. 6). The control rules out artifacts: the same detector and windowing applied to the $A = 0.2$ never-absorbers over a median of $\sim$200 cycles gives $\langle \Delta M \rangle \approx -0.001$ and a ratchet fraction of exactly zero — stationary, with a slight *relaxation* consistent with settling onto an attractor.

5.3 What collapse is not, and what organizes it

Because the oscillators are strictly identical, a natural mechanism would be escape transverse to the Poisson (Ott–Antonsen) submanifold [13, 14] on which the reduced chimera solutions live. We tested this directly with the moment observable $D = \sum_{m=2}^{4} |Z_m - Z_1^m|^2$, validated in three stages (the Poisson identity $Z_m = Z_1^m$ to machine precision on Möbius-constructed states; dynamical invariance of on-manifold initial conditions under the shipped integrator; a clustered-state contrast), and the hypothesis fails comprehensively: initial distance D_0 does not predict lifetime ($|\rho| \leq 0.15$ at all fourteen parameter points), event-aligned $\langle D \rangle$ shows no ramp preceding collapse, and a paired interventional experiment injecting controlled D_0 levels leaves lifetimes flat. Collapse is not transverse escape.

It is instead organized in the collective coordinates. Among static features of the seed, the initial inter-population phase offset dominates: $|\Delta\phi_0|$ anticorrelates with lifetime at $\rho \approx -0.43$ to -0.59 for $N \geq 8$ (median magnitude 0.52), versus the D_0 null's ≤ 0.08, and a cross-validated regression on the $t = 0$ collective features reaches $R^2 \approx 0.2$ (quadratic up to 0.44). Larger initial phase separation between the populations means earlier death — a fact Sec. 6 converts from regression to dynamics.

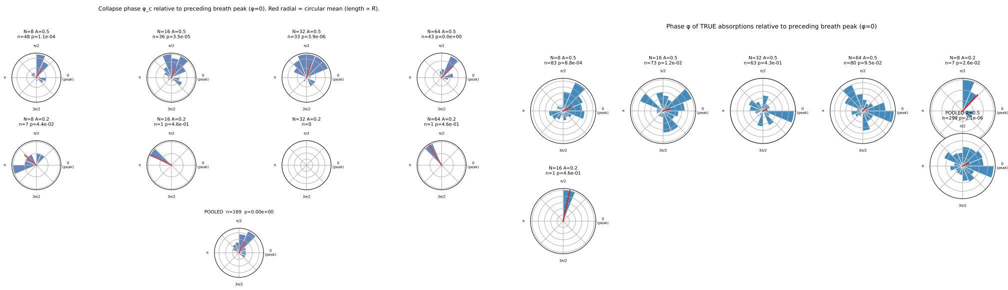

Figure 5: Breath-phase locking. Circular histograms of collapse-attempt phases (left) and true-absorption phases (right) relative to the preceding breathing-envelope peak, $A = 0.5$. Attempts cluster at 0.13–0.20 of the cycle (pooled Rayleigh $p \approx 0$); absorptions inherit the lock (pooled $p = 2.2 \times 10^{-6}$).

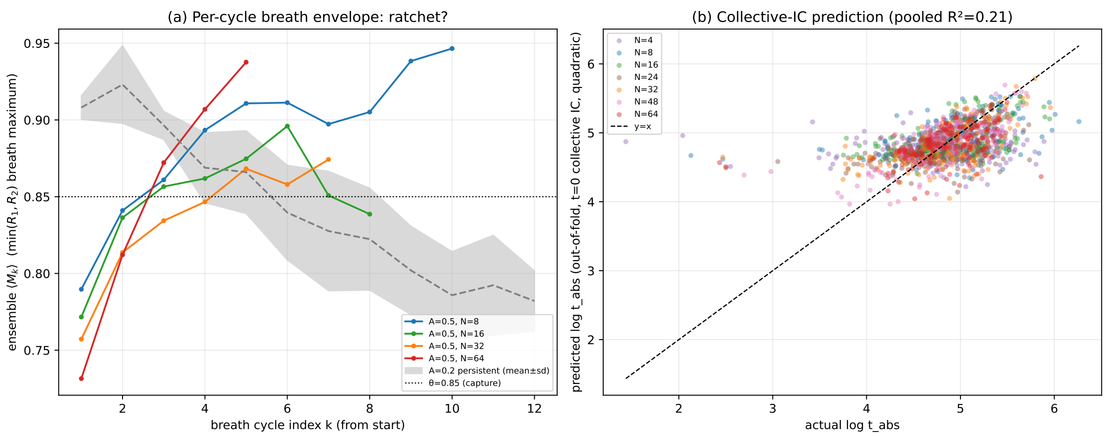

Figure 6: The ratchet. Ensemble-averaged breathing maxima $\langle M_k \rangle$ versus cycle index k at $A = 0.5$ for several N, with the $A = 0.2$ never-absorber band (stationary, $\langle \Delta M \rangle \approx -0.001$) for contrast. Per-cycle increments are strictly positive at every N; $\log(\theta - M_k)$ falls linearly (median slopes -0.55 to -0.77).

6 Reduced dynamics: the post-homoclinic ghost

6.1 The reduced system, validated against its source

In the $N \to \infty$ mean-field, the two-population system closes in the per-population order parameters. Writing $\rho_\sigma e^{-i\phi_\sigma}$ for population σ's centroid, the reduced flow is [3]

$$\dot{\rho}_1 = -\frac{\rho_1^2 - 1}{2}\Big[\mu\rho_1 \cos\alpha + \nu\rho_2 \cos(\phi_2 - \phi_1 - \alpha)\Big], \tag{2}$$

together with the corresponding $\dot{\phi}_1$ equation and the $1 \leftrightarrow 2$ images; rotational symmetry reduces this to three variables $(\rho_1, \rho_2, \psi = \phi_1 - \phi_2)$. On the invariant manifold $\rho_1 = 1$ (synchronized population exactly locked) the flow further reduces to two variables $(r = \rho_2, \psi)$, Eq. (12) of Ref. [3], whose stationary chimeras, saddle-node curve $A_{SN}(\beta)$, and Hopf curve $A_H(\beta)$ are known in closed or series form, with the linear stability of these two-population chimeras characterized in detail by Laing [15].

Before applying any of this to our system, the implementation was gated against the source paper itself: the parametrized stationary chimeras zero the reduced right-hand side to 4×10^{-16}; integration reproduces the published regimes at $\beta = 0.1$ (stationary at $A = 0.2$; breathing at $A = 0.28$ and 0.35, with the longer

Table 2: Reduced flow versus finite-N measurement at the operating corner ($A = 0.5$, $\beta = 0.05$). No fitted parameters; times in seconds at the integrator's documented 1:1 clock. The ratio is reduced/measured throughout; the final row reports the $A = 0.2$ chimera level for comparison.

quantity	reduced	measured	ratio (reduced/measured)
breath period T_b	25.0 (linear 19.2)	21–25	≈ 1.1
per-cycle spiral slope (pooled)	-0.67	-0.55 to -0.77	≈ 1.0
median capture time	43	139	0.31
A=0.2: chimera level	$r^* = 0.678$	$\overline{R}_{\mathrm{incoh}} = 0.66$–$0.68$	≈ 1.0

period at the larger A); and numerically located saddle-node and Hopf points match the published series to 5×10^{-5}. Every number below is parameter-free and, by the integrator's documented clock, in literal seconds.

6.2 The two corners straddle the bifurcation set

At $\beta = 0.05$: $A_{SN} = 0.0948$, $A_H = 0.2703$, and the homoclinic — located by bisection on the divergence of the breathing cycle's period — at $A_{hc} = 0.4096$ (bracket width 8.8×10^{-5}). Traced over $\beta \in [0.03, 0.20]$, the homoclinic curve descends, turns, and rises into the Takens–Bogdanov point at $(0.2239, 0.3372)$ where it meets the saddle-node and Hopf curves — the non-monotone approach is demanded by the termination and serves as an internal consistency check (Fig. 7).

The verdicts follow immediately. $A = 0.2$ lies in the stable-chimera band ($A_{SN} < 0.2 < A_H$): the reduced system has a stable stationary chimera at $r^* = 0.6781$ (integration amplitude decaying to 10^{-9}), with weak relaxation $\sigma = -0.0054\,\mathrm{s}^{-1}$ — matching the never-absorbers, whose mean incoherent level sits at r^* and whose envelope drifts slightly *down*, and explaining the bimodal capture as a basin question: seeds either fall to synchrony on the first approach or onto the chimera attractor forever. $A = 0.5$ lies beyond the homoclinic ($0.5 > A_{hc} = 0.4096$): *no chimera attractor exists at the operating corner*. The chimera fixed point is an unstable spiral ($\sigma = +0.01243\,\mathrm{s}^{-1}$, $\omega = 0.3277\,\mathrm{s}^{-1}$), the breathing limit cycle that once surrounded it is destroyed, and sync is globally attracting. What the deployed voice has been operating is a transient spiraling outward along the ghost of that destroyed cycle (Fig. 8).

6.3 Quantitative match with no fitted parameters

Table 2 compares the reduced flow to the finite-N measurements. The spiral's rotation gives a linear breath period $2\pi/\omega = 19.2\,\mathrm{s}$ and a nonlinear period of $25.0\,\mathrm{s}$ for trajectories at canonical-seed amplitudes — bracketing the measured 21–25 s. The reduced per-cycle approach slopes, computed identically to the measurement from each run's seed-mapped initial condition, are $-0.62, -0.66, -0.66, -0.73$ for $N = 8$–64 against measured $-0.55, -0.50, -0.59, -0.77$: the trend with N is reproduced even though the reduced flow is N-free, because the seed ensembles at different N occupy different collective initial conditions (mean $R_{\mathrm{incoh},0}$ falls $\sim 1/\sqrt{N}$). The single substantial mismatch is the absolute capture time: the reduced median is 43 s against the measured 139 s — the mean-field flow dies $3.2\times$ faster than the finite-N system (Sec. 7). (A single trajectory started near the focus, by contrast, escapes only at $\approx 726\,\mathrm{s}$: at $\sigma = 0.0124\,\mathrm{s}^{-1}$ the $\sim$9 e-folds from a near-focus deviation take exactly that long, while the seed ensemble starts at finite amplitude — the two timescales are the same spiral read from different starting radii.)

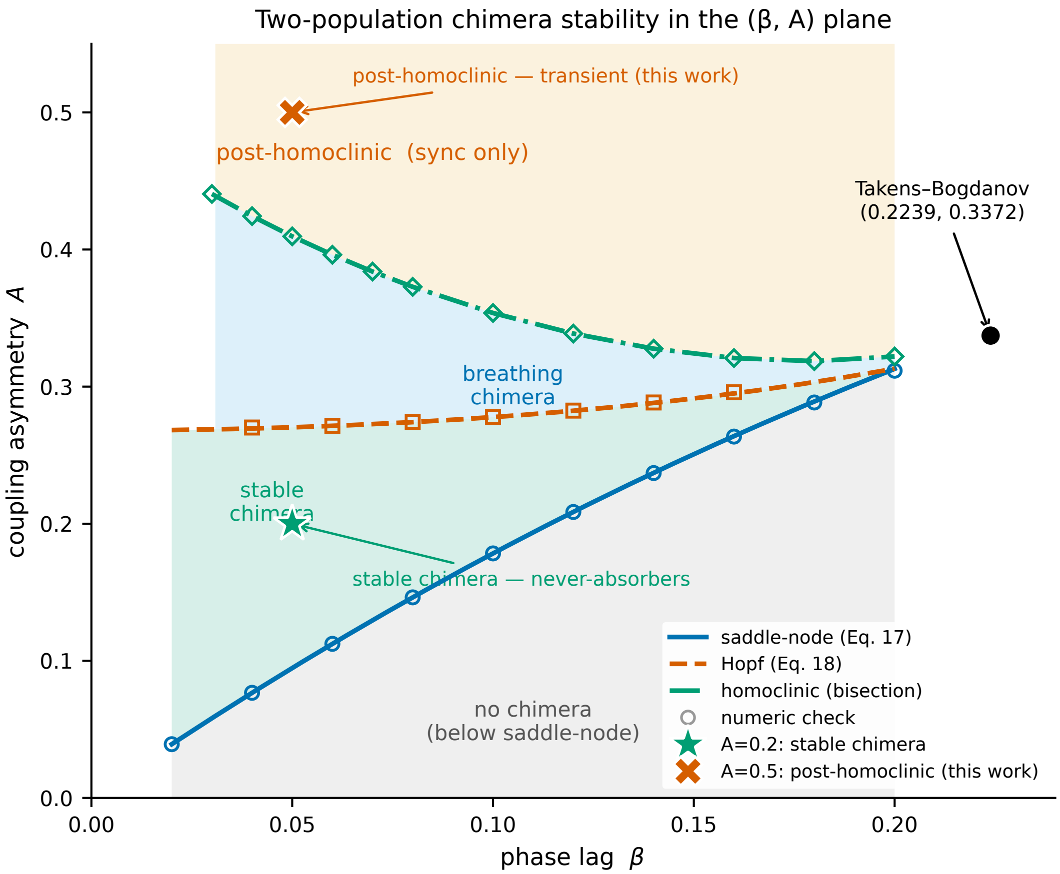

Figure 7: Stability diagram of the two-population system in the (β, A) plane: saddle-node (Eq. 17 of Ref. [3]) and Hopf (Eq. 18) curves with numeric checks, the numerically traced homoclinic curve terminating at the Takens–Bogdanov point $(0.2239, 0.3372)$, and the two operating corners of the deployed system. The nominally stable corner ($A = 0.5$) is post-homoclinic — no chimera attractor — while the nominally unreliable corner ($A = 0.2$) sits inside the stable-chimera band.

6.4 Individual lifetimes from a three-variable model

Finally, the reduced system predicts *individual* deaths. Mapping every $A = 0.5$ seed to its collective coordinates $(\rho_1, \rho_2, \psi)_0 = (R_{\text{sync}}, R_{\text{incoh}}, \Delta\phi)_0$ and integrating the three-variable flow to capture yields a predicted lifetime whose Spearman correlation with the measured t_{abs} is 0.445 pooled (up to 0.60 per N) — comparable to the best fitted static feature — and, crucially, carries a *positive partial correlation of* $+0.20$–0.32 *beyond* $|\Delta\phi_0|$ *at every* N: the parameter-free dynamics predicts lifetime variance no static regression on the same inputs captures (Fig. 9). The transverse coordinate stays quiet throughout (ρ_1 deviating from 1 by ≤ 0.003 median), justifying the manifold picture a posteriori. This N-independence is not merely a mean-field artifact: for identical oscillators under global sinusoidal coupling, Watanabe–Strogatz theory [18, 19] reduces each population to a three-parameter flow at *every* N, with the remaining $N-3$ degrees of freedom frozen as constants of motion — as the group-theoretic [17] and hierarchical-population [16] formulations make explicit. The three-variable system integrated here is exact on the uniform-constants (Poisson) submanifold, where the collective dynamics closes independently of N; the off-manifold constants, which a finite-N ensemble need not realize, are precisely where the residual $3.2\times$ prolongation must originate — sharpening, not softening, the open problem.

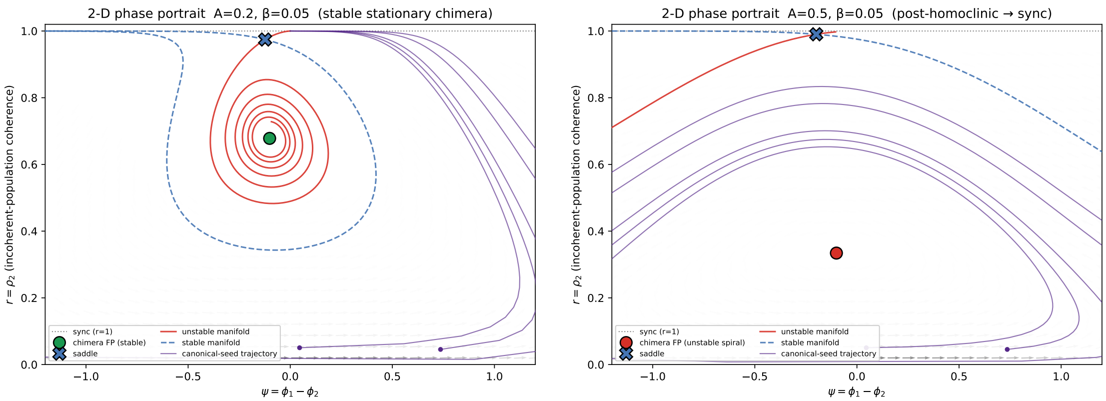

Figure 8: Reduced-system phase portraits at $\beta = 0.05$. Left ($A = 0.2$): stable chimera spiral at $r^* = 0.678$ with the saddle and its manifolds organizing the basin. Right ($A = 0.5$): the chimera fixed point is an unstable spiral ($\sigma = +0.0124\,\mathrm{s}^{-1}$); canonical-seed trajectories spiral out to synchronization.

7 Discussion

The inversion, operationally. The deployed voice's celebrated reliability — $\approx 96\%$ in-state over half-hour renders — is revealed as 100% supervisor-maintained: at the operating corner there is nothing to be reliable, since every unsupervised trajectory dies in $\approx 139\,\mathrm{s}$ regardless of how many oscillators are added. Conversely the corner discarded as unreliable holds the system's only genuine attractor. For applications, two consequences follow. First, supervision is *intrinsic* to post-homoclinic operation at any N — re-seeding is not a finite-size patch but the entire persistence mechanism. Second, collapse detection should be absorption-grade: a detector with a recovery window (rather than the 2-s sustain rule, which over-triggers on 69–97% of firings) would intervene an order of magnitude less often for the same uptime.

The open problem: noise that prolongs. The reduced flow explains the geometry of death — where the corners sit, the breath period, the spiral rate, even individual lifetimes' ordering — but captures $3.2\times$ too fast, and the flat $\tau_{\mathrm{abs}}(N)$ plateau has no analog in an N-free collective flow. Both point the same direction: finite-N fluctuations *extend* the transient's life relative to mean-field, the opposite of the usual intuition in which noise destroys metastable states. At the operating corner this prolongation is N-independent — the same factor of 3.2 across the full $16\times$ range of N — though that independence is itself a property of the corner's depth beyond the homoclinic, not of every post-homoclinic parameter: at $\beta = 0.10$ it survives at the depth-matched point (coefficient of variation 0.14 across N) but weakens at the deeper $A = 0.5$, where the factor falls from 2.0 to 1.1 (Appendix A). A natural candidate for the prolongation is fluctuation-induced re-injection — finite-size kicks scattering the trajectory back toward the ghost region the deterministic spiral is leaving. We tested the simplest version directly (Appendix B): additive Gaussian noise of amplitude $c/\sqrt{N}$ on the collective variables of the reduced flow. It does produce re-injection and prolongation, but the prolongation is N-*dependent* — strong at small N and absent by $N = 64$ — because the noise amplitude is. The observed N-independent factor of 3.2 at the operating corner therefore rules out additive collective noise and points to a mechanism whose effective re-injection strength does not scale away as $1/\sqrt{N}$ (for example structured or multiplicative fluctuations tied to the breath cycle). We report the puzzle as measured and leave the responsible mechanism open; the rising-hazard (aging-in-cycles) structure and the breath-locked phase distribution remain sharp constraints any such theory must satisfy.

Relation to ring-topology transients. On nonlocally coupled rings, finite-size chimeras are chaotic transients with lifetimes growing rapidly in N [4]. The two-population mean-field case is qualitatively different: its collective flow is N-free, the N-dependence entering only through the initial-condition ensemble

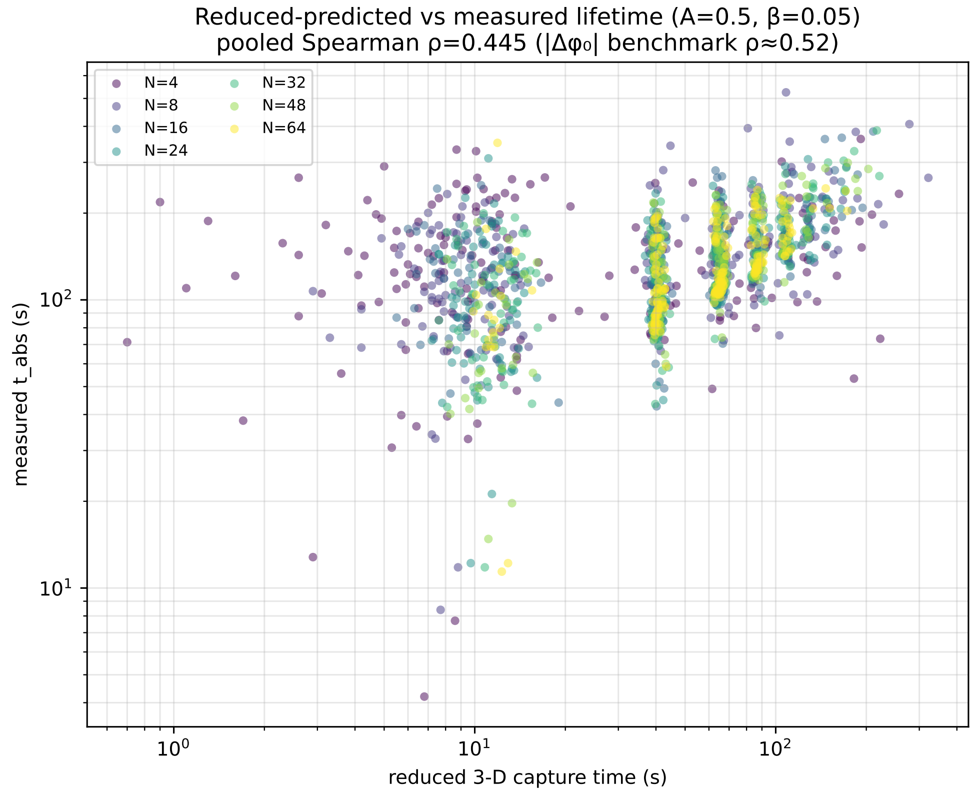

Figure 9: Run-by-run lifetime prediction. Reduced-model capture time from each seed's collective initial condition versus measured t_{abs}, colored by N ($A = 0.5$; pooled Spearman $\rho = 0.445$; partial correlation beyond $|\Delta\phi_0|$ positive at every N).

— and correspondingly we find no lifetime growth at all. The contrast suggests the N-scaling of chimera transience is a property of coupling topology rather than of chimeras per se.

Scope. All results are at strictly identical frequencies, equal population sizes, the shipped phase lag ($\beta = 0.05$, with the phenomenology reproduced at $\beta = 0.10$; Appendix A), and one canonical seed family; the absolute τ_{abs} carries the documented 22–27% labeling sensitivity; and the collective-coordinate summaries are a lossy projection of the $2N$-phase state (weakest at $N = 4$, where they predict little). Within that scope, every headline claim is criterion-robust, bit-reproducible, and regenerable from committed configurations.

8 Conclusion

An engineering question — why does a synthesizer's chimera need so much re-seeding? — unwound into a dynamical one. The collapse detector turns out to detect grazes, not deaths: most sustained near-synchrony excursions heal, and true absorption takes twice as long as first passage suggests. Corrected lifetimes are N-independent and aging, deaths lock to the breathing phase and arrive by a monotone ratchet of the envelope, and the bifurcation structure of the reduced two-population model explains why: the operating corner lies beyond the homoclinic, so the deployed state is a transient on a destroyed cycle's ghost, while the discarded corner holds the true chimera attractor. A three-variable, parameter-free flow predicts the period, the spiral rate, and the ordering of individual deaths — and underestimates their timing by $3.2\times$, leaving one clean open problem: at finite N, fluctuations make this chimera live longer, not die faster.

Author Note

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Author Declarations

Conflict of Interest

The author has no conflicts to disclose.

Author Contributions

Avery Karlin: Conceptualization (lead); Methodology (lead); Software (lead); Formal analysis (lead); Investigation (lead); Data curation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead).

Data Availability

The data and code that support the findings of this study — the integrator, experiment harnesses, campaign records (JSONL), committed configurations, and figure-generation scripts — are openly available in the annealMusic repository at github.com/akarlin3/annealMusic and archived at Zenodo (doi.org/10.5281/zenodo.20564867); python3 tools/paper-figures/run_all.py regenerates all figures.

A Robustness of the phenomenology at $\beta = 0.10$

To show the operating-corner phenomenology is not an artifact of the shipped phase lag $\beta = 0.05$, we repeated the finite-N campaign at $\beta = 0.10$, the value at which the reduced model is validated against the published regimes [3]. Reduced-model placement gives a saddle-node at $A_{SN} = 0.178$, a Hopf at $A_H = 0.278$, and a homoclinic at $A_{hc}(0.10) = 0.354$ (bisection bracket width 9×10^{-5}). We sampled the post-homoclinic corner $A = 0.5$ (parallel to the operating corner) and, to match its distance beyond the homoclinic, a depth-matched point $A = A_{hc} + 0.090 = 0.444$; $A = 0.2$ and a cleaner mid-band point $A = 0.24$ sit in the stable-chimera band as never-absorber controls.

Every headline result survives (Fig. 10). The absorption lifetime is flat in N ($\hat{\tau}_{\text{abs}}(64)/\hat{\tau}_{\text{abs}}(8) = 0.93$ at $A = 0.5$; 0.87 at $A = 0.444$; fitted exponential rate $\approx -1 \times 10^{-3}$ per oscillator, zero censoring). The censored-Weibull shape stays above one ($k_{\text{abs}} = 1.8$–2.8 at $A = 0.5$, 1.4–1.6 at $A = 0.444$), so hazard still rises. The per-cycle ratchet is positive ($\langle \Delta M \rangle = +0.06$ to $+0.19$, bootstrap CIs excluding zero), and true absorptions remain breath-phase-locked (pooled Rayleigh $p = 6.5 \times 10^{-5}$). The reduced flow reproduces the breath period (≈ 23–25 s, measured 20–25 s) and spiral rate. The stable-band controls behave as expected:

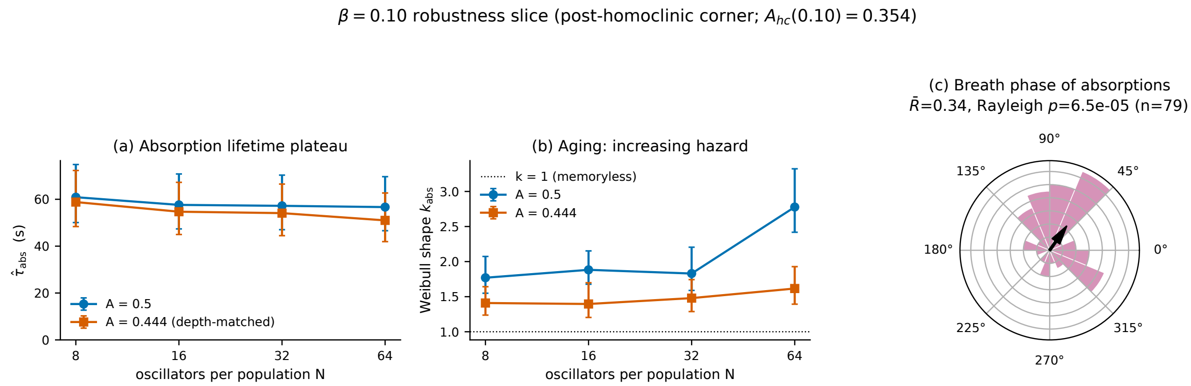

Figure 10: Robustness at $\beta = 0.10$. (a) Absorption lifetime $\hat{\tau}_{\text{abs}}(N)$ is flat in N at both the post-homoclinic corner $A = 0.5$ and the depth-matched $A = 0.444$. (b) Censored-Weibull shape $k_{\text{abs}}(N)$ stays above the memoryless value $k = 1$ (aging persists). (c) True-absorption breath phases remain clustered (pooled $\bar{R} = 0.34$, Rayleigh $p = 6.5 \times 10^{-5}$, $n = 79$).

the mid-band $A = 0.24$ holds a near-N-independent persistent fraction (44% at $N = 8$, 52% at $N = 64$; 50 seeds each), a cleaner never-absorber control than $A = 0.2$, whose persistent fraction swings more widely (28% to 60%) as it sits close to A_{SN}.

One caveat refines a body-text claim. The transient is shallower at $\beta = 0.10$ (≈ 2 breath cycles to absorption vs ≈ 6 at the operating corner), and correspondingly the reduced-to-measured capture ratio — the $3.2\times$ prolongation of Sec. 7 — is N-independent only at comparable post-homoclinic depth. At the depth-matched $A = 0.444$ it stays flat (coefficient of variation 0.14 across N), but at the deeper $A = 0.5$ it falls from 2.0 to 1.1 across N. The N-independence of the prolongation is thus a property of the operating corner's depth beyond the homoclinic, not of every post-homoclinic parameter.

B A finite-size noise test of fluctuation re-injection

The reduced flow captures $\approx 3.2\times$ faster than the finite-N system, and at the operating corner that prolongation is N-independent (Sec. 7). The natural candidate — fluctuation-induced re-injection — predicts that adding finite-size noise to the collective flow should reproduce both facts. We tested this directly, integrating the three-variable reduced flow at the operating corner ($A = 0.5$, $\beta = 0.05$) with the Euler–Maruyama scheme and adding Gaussian noise of amplitude $\sigma_{\text{noise}} = c/\sqrt{N}$ to the collective variables (ρ_1, ρ_2, ψ), with reflection keeping ρ in $[0, 1]$. At $c = 0$ the scheme reproduces the deterministic capture times to a median 0.25% per initial condition. The physical scale, estimated from the measured high-frequency fluctuation of R_{incoh} in the baseline data, is $c \approx 0.05$. We swept $N \in \{8, 16, 32, 64\}$, $c \in \{0, 0.025, 0.05, 0.1, 0.2\}$, 200 realizations each, from the Sec. 6 seed-mapped initial conditions.

The result is a clean negative (Fig. 11). At the physical scale $c \approx 0.05$ the noise barely perturbs capture (prolongation factor ≈ 0.97, N-independent). Larger amplitudes do prolong — at $c = 0.2$ the trajectory is repeatedly re-injected, accumulating up to $\sim$240 breath cycles before capture — but the prolongation is strongly *N-dependent* (factor 2.4 at $N = 8$ falling to 0.7 at $N = 64$; coefficient of variation 0.40 across N), and never reaches the measured ≈ 3.2 at any cell where it is N-independent. The rising-hazard constraint holds only at small c ($k > 1$ for $c \leq 0.05$, $k \approx 1$ for $c \geq 0.1$), while breath-phase locking survives throughout. Thus additive $1/\sqrt{N}$ noise on the collective flow reproduces the mechanism (re-injection) but with the wrong N-scaling: because its amplitude is N-dependent, so is the prolongation it produces. The N-independent $3.2\times$ factor requires a different mechanism.

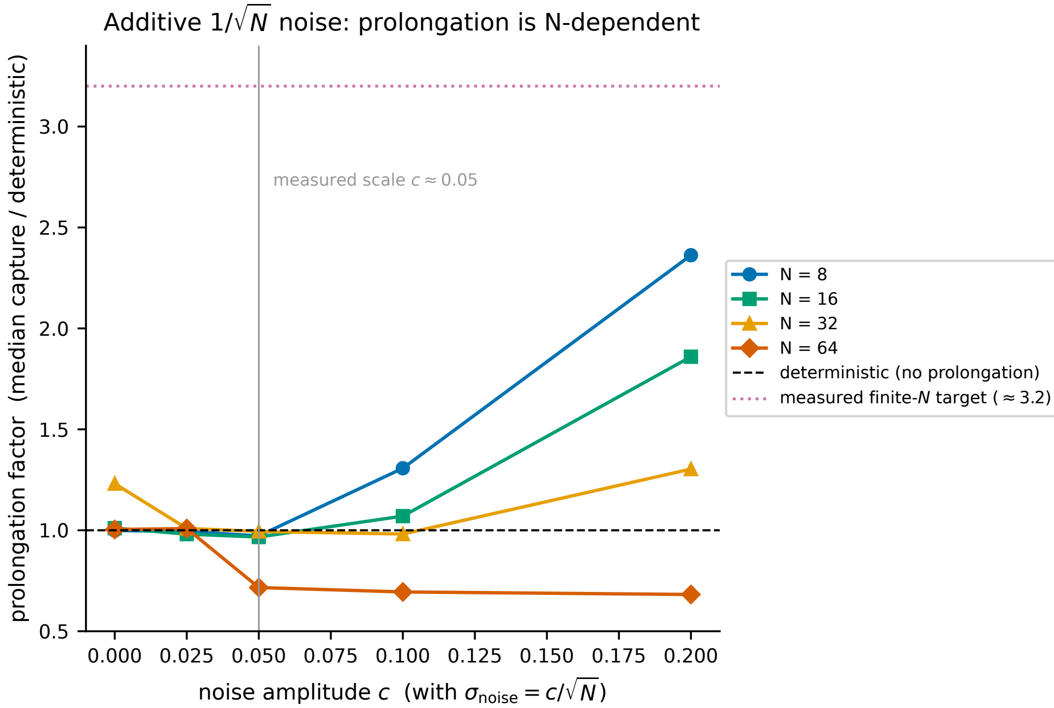

Figure 11: Finite-size noise test. Prolongation factor (median capture relative to the deterministic reduced flow) versus additive-noise amplitude c ($\sigma_{\text{noise}} = c/\sqrt{N}$) at $A = 0.5$, $\beta = 0.05$. At the physical scale $c \approx 0.05$ the curves sit near unity; at larger c they fan out with N, so the prolongation is N-dependent and never reaches the measured ≈ 3.2 (dotted) where it is N-independent.

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